

The empirical recurrence for OEIS A296639: recovering a conjecture that is not written down

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Abstract

OEIS A296639 records “Empirical recurrence of order 74”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 107$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 74, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 74 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A296639 is “Number of $n \times 4$ 0..1 arrays with each 1 horizontally, vertically or antidiagonally adjacent to 1, 2 or 4 neighboring 1s.”. It has offset 1 and begins

7, 86, 671, 5696, 49872, 430522, 3713779, 32095222, ...

The entry states:

Empirical recurrence of order 74 (see link above)

As of the “Last modified” line on the live entry (Dec 17 2017 15:09:31, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 107$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 107$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 214$ exact terms of the model returns order 74 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{74} c_i a(n-i),$$

c_1	13	c_{20}	8202391	c_{39}	73770388	c_{58}	−162755
c_2	−48	c_{21}	−4707046	c_{40}	−100690368	c_{59}	249681
c_3	137	c_{22}	−2979074	c_{41}	74773620	c_{60}	−134229
c_4	−553	c_{23}	7090996	c_{42}	−82256804	c_{61}	127209
c_5	1437	c_{24}	−329887	c_{43}	54974837	c_{62}	−89373
c_6	−2685	c_{25}	−14706176	c_{44}	−58066851	c_{63}	110304
c_7	8798	c_{26}	27166539	c_{45}	44233104	c_{64}	−64673
c_8	−27633	c_{27}	−24725223	c_{46}	−47299530	c_{65}	53541
c_9	60265	c_{28}	3346662	c_{47}	36935371	c_{66}	−35172
c_{10}	−135502	c_{29}	20568498	c_{48}	−33601795	c_{67}	18090
c_{11}	284830	c_{30}	−27095768	c_{49}	22007837	c_{68}	−10500
c_{12}	−291094	c_{31}	19892457	c_{50}	−15820622	c_{69}	4666
c_{13}	−170984	c_{32}	321245	c_{51}	9790262	c_{70}	−1668
c_{14}	642875	c_{33}	11701825	c_{52}	−5330210	c_{71}	549
c_{15}	106405	c_{34}	−20359157	c_{53}	3389438	c_{72}	−152
c_{16}	−2156491	c_{35}	40951990	c_{54}	−2015303	c_{73}	24
c_{17}	2998108	c_{36}	−57799352	c_{55}	1310454	c_{74}	−1
c_{18}	166558	c_{37}	52561481	c_{56}	−611295		
c_{19}	−5726761	c_{38}	−83945914	c_{57}	721625		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 74, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $74 + 4$ terms removed returns order 74 as well, so $\deg q_{\min} = 74 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $74 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 74 that this sequence satisfies — and that family turns out to have exactly one member.

References

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